Daniël Trujillo

(857) 209-6069 | danieltr@mit.edu | linkedin.com/in/trujillo-daniel

Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
PhD in EECS, CSAIL. Advisor: Mengjia Yan — *Sept. 2023 – Expected end of 2027*

ETH Zürich — Zürich, Switzerland
*MSc in Computer Science. GPA 5.49 / 6.00 (**ETH Medal**)* — *Sept. 2020 – June 2023*

Vrije Universiteit (VU) Amsterdam — Amsterdam, Netherlands
*BSc in Computer Science. GPA: 9.3 / 10.0 (**Cum Laude**)* — *Sept. 2017 – Aug. 2020*

Publications *(* = joint first authorship)*

TONTOU: On the Exploitability of Time-of-Neutralization to Time-of-Use Windows *(Distinguished Paper Award Runner-Up)* — Aug. 2026
Daniël Trujillo, Mengjia Yan
USENIX Security 2026

Defeating Transient Execution Attacks by Limiting Secret Reachability through Register Hiding and ShadowCFI — May 2026
Daniël Trujillo, Jagadish Kotra, David Kaplan, Mengjia Yan
IEEE S&P 2026

Phantom: Exploiting Decoder-detectable Mispredictions *(Best Paper Award)* — Oct. 2023
Johannes Wikner*, Daniël Trujillo*, Kaveh Razavi
MICRO 2023

Inception: Exposing New Attack Surfaces with Training in Transient Execution — Aug. 2023
Daniël Trujillo*, Johannes Wikner*, Kaveh Razavi
USENIX Security 2023

TLB;DR: Enhancing TLB-based Attacks with TLB Desynchronized Reverse Engineering — Aug. 2022
Andrei Tatar*, Daniël Trujillo*, Cristiano Giuffrida, Herbert Bos
USENIX Security 2022

Experience

NVIDIA — May 2026 – Aug. 2026
PhD Research Intern, Security and Privacy — *Westford, MA, USA*
- Designed and implemented an AI-assisted pipeline for assessing the exploitability of Spectre gadgets
- Demonstrated exploitability of two previously unknown Spectre v1 gadgets in the Linux kernel

Microsoft Research — June 2025 – Aug. 2025
Research Intern, Software HARdware Co-design (SHARC) — *Redmond, WA, USA*
- Re-programmed memory controller to enable unconventional physical-to-DRAM mappings
- Enabled cache-isolated memory sharing, variable page-sizes (e.g., 1MB pages) and more on today's x86 hardware

AMD — June 2024 – Jan. 2025
Research Intern, Research and Advanced Development (RAD) — *Austin, TX, USA*
- Researched more performant mitigations against transient execution attacks
- Software-only Spectre v2 defense for the Linux kernel, reducing overhead from 33.4% to 25.8% on server workloads

Intel Corporation — July 2022 – Nov. 2022
Offensive Security Research Intern, Intel Product Assurance and Security (IPAS) — *Zürich, Switzerland*
- Contributed to taint analysis prototype, specialized in finding stale data bugs semi-automatically
- Demonstrated existence of a known stale bug in Intel IP

IBM Research — June 2021 – Sep. 2021
Intern, Cloud Data Platforms — *Zürich, Switzerland*
- Developed PoC of software abstraction of smartNIC storage-offloading functionalities

CVEs

CVE-2026-68480 (TONTOU) — Aug. 2026
Linux

CVE-2023-20569 (Inception / "SRSO") — Aug. 2023
AMD

CVE-2022-23825 (Phantom / "Branch Type Confusion") — Oct. 2022
AMD

Awards & Honors

Distinguished Paper Award Runner-Up — August 2026
USENIX Security

Intel Bug Bounty Program Bonus — March 2026
Intel Corporation

NDSEG Fellowship — Sep. 2024 – Aug. 2027
U.S. Department of Defense

Best Paper Award — Oct. 2023
MICRO 2023

ETH Medal — Sep. 2023
ETH Zürich

Best Hack & Use of Google Cloud — Oct. 2019
HackHarvard

GLOBE Scholarship — Sep. 2019
VU Amsterdam

Patents

Enabling secure memory sharing on cloud servers — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

Custom-sized pages for reducing TLB misses — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

Fine-grained cache hit/miss profiling — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

System and method for preventing transient execution attacks — Jun. 2025
Jagadish Kotra, Daniël Trujillo, David Kaplan — *(Pending)*
AMD

Detecting potential security issues in a hardware design using RTL information flow tracking — Mar. 2023
Ben Gras, Daniël Trujillo — *(Pending)*
Intel Corporation

Selected Talks

TONTOU: Weaponizing Interrupts to Break Spectre Defenses on AMD and Intel — Aug. 2026
Daniël Trujillo — *Las Vegas, NV, USA*
Black Hat USA

From Phantom to Inception — May 2024
Daniël Trujillo, Johannes Wikner — *Berlin, Germany*
OffensiveCon

Inception — Dec. 2023
Daniël Trujillo — *São Paulo, Brazil*
H2HC

Teaching Experience

ETH Zürich — Jan. – May 2022
Teaching Assistant — Computer Engineering — *Zürich, Switzerland*

VU Amsterdam — June 2018 – May 2020
Teaching Assistant — *Amsterdam, Netherlands*

- Programming for Economists in Python (Jun 2018 & Jun 2019)
- Computer Programming in C++ (Sept – Oct 2018)
- Physical Computing (Nov – Dec 2018)
- Web Technology (Jan 2019 & Jan 2020)
- Pervasive Computing (Feb – Mar 2019)
- Natural Language Processing Technology (Apr – May 2020)